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CS 385 Algorithms
Professor Eiers
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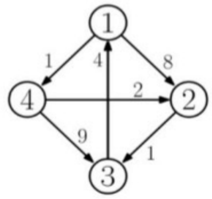
"I pledge my honor that I have abided by the Stevens Honor System"
└ Arjun Patel

1. Apply the Floyd-Warshall Algorithm to find the all-pair shortest paths for the following graph. State the shortest paths for each vertex to every other vertex. (10 points)



Recurrence Relation

$$d_{ij}^{(k)} = \begin{cases} W_{ij} & k = 0 \\ \min\left(d_{ij}^{(k-1)},\ d_{ik}^{(k-1)} + d_{kj}^{(k-1)}\right) & k \geq 1 \end{cases}$$

$$\pi_{ij}^{(0)} = \begin{cases} \text{NIL} & \text{if } i=j \text{ or } W_{ij} = \infty \\ i & \text{if } i \neq j \text{ AND } W_{ij} \neq \infty \end{cases}$$

$$D^{(0)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & \infty & 1 \\ 2 & \infty & 0 & 1 & \infty \\ 3 & 4 & \infty & 0 & \infty \\ 4 & \infty & 2 & 9 & 0 \end{array} \qquad \Pi^{(0)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & & & & \\ 2 & & & & \\ 3 & & & & \\ 4 & & & & \end{array}$$

$$D^{(1)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & \infty & 1 \\ 2 & \infty & 0 & 1 & \infty \\ 3 & 4 & 12 & 0 & 5 \\ 4 & \infty & 2 & 9 & 0 \end{array} \qquad \Pi^{(1)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & & & & \\ 2 & & & & \\ 3 & & 1 & & 1 \\ 4 & & & & \end{array}$$

$D^0[2,3] \quad D^0[2,1] + D^0[1,3]$
$1 < \infty + \infty$

$D^0[2,4] \quad D^0[2,1] + D^0[2,4]$
$\infty = \infty + \infty$

$D^0[3,2] \quad D^0[3,1] + D^0[1,2]$
$\infty > 4 + 8$

$D^0[3,4] \quad D^0[3,1] + D^0[1,4]$
$\infty > 4 \quad 1$

$D^0[4,2] \quad D^0[4,1] + D^0[1,2]$
$2 < \infty + 8$

$D^0[4,3] \quad D^0[4,1] + D^0[1,3]$
$9 < \infty + \infty$

we repeat this process until we cover / upate all shortest paths

$$D^{(2)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & 9 & 1 \\ 2 & \infty & 0 & 1 & \infty \\ 3 & 4 & 12 & 0 & 5 \\ 4 & \infty & 2 & 3 & 0 \end{array} \qquad \Pi^{(2)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & & & 2 & \\ 2 & & & & \\ 3 & & 1 & & 1 \\ 4 & & & 2 & \end{array}$$

$$D^{(3)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & 9 & 1 \\ 2 & 5 & 0 & 1 & 6 \\ 3 & 4 & 12 & 0 & 5 \\ 4 & 7 & 2 & 3 & 0 \end{array} \qquad \Pi^{(3)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & & & 2 & \\ 2 & 3 & & & 3 \\ 3 & & 1 & & 1 \\ 4 & 3 & & 2 & \end{array}$$

$$\boxed{D^{(4)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & 4 & 1 \\ 2 & 5 & 0 & 1 & 6 \\ 3 & 4 & 7 & 0 & 5 \\ 4 & 7 & 2 & 3 & 0 \end{array} \qquad \Pi^{(4)} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & & 4 & 4 & \\ 2 & 3 & & & 3 \\ 3 & & 4 & & 1 \\ 4 & 3 & & 2 & \end{array}}$$

shortest paths

1 → 2 : 3	4 → 1 : 7
1 → 3 : 4	4 → 2 : 2
1 → 4 : 1	4 → 3 : 3
2 → 1 : 5	
2 → 3 : 1	
2 → 4 : 6	
3 → 1 : 4	
3 → 2 : 7	
3 → 4 : 5	

1 → 4 → 2
1 → 4 → 2 → 3
1 → 4
1 → 1
2 → 2
3 → 3
4 → 4

4 → 2 → 3 → 1
2 → 3 → 1
2 → 3
2 → 3 → 1 → 4
3 → 1
3 → 1 → 4 → 2
3 → 1 → 4
4 → 2 → 3
4 → 2

3. Given the following coins, fill out the following table for finding the maximum amount of money subject to the constraint that no two adjacent coins can be picked up. State which coins were picked up to compute the maximum value. (10 points)

Index i	0	1	2	3	4	5	6
C_i	0	3	5	2	6	4	1
F(i)							
S(i)							

index i	0	1	2	3	4	5	6
C_i	0	3	5	2	6	4	1
F(i)	0	3	5	5	11	11	12
S(i)	0	0	0	2	2	4	4

Base-cases

$F(0) = 0$ no coins at all

$F(1) = C_1$ only one coin so we pick it up

Recurrence Relation:

$$F(n) = \max(\underbrace{C_n + F(n-2)}_{\text{with } C_n}, \underbrace{F(n-1)}_{\text{without } C_n}), \text{ for } n > 1$$

$$S(i) = \begin{cases} i-2 & \text{if } F(i) = C_i + F(i-2) \quad \text{pick up coin} \\ i-1 & \text{if } F(i) = F(i-1) \quad \text{do not pick up coin} \end{cases}$$

$F(2) = \max(5 + F(0), F(1)) = \max(5+0, 3) = 5$
$S(i) = i-2 = 2-2 = 0$

$F(3) = \max(2 + F(1), F(2)) = \max(2+3, 5) = 5$
$S(i) = i-1 = 3-1 = 2$

$F(4) = \max(6 + F(2), F(3)) = \max(6+5, 5) = 11$
$S(i) = i-2 = 4-2 = 2$

$F(5) = \max(4 + F(3), F(4)) = \max(4+5, 11) = 11$
$S(i) = i-1 = 4$

$F(6) = \max(1 + F(4), F(5)) = \max(1+11, 11) = 12$
$S(i) = i-2 = 6-2 = 4$

Now, we backtrack to find the coins we picked up:

- if $F(i) > F(i-1)$, then we picked up coin c_i and we go back to $S(i) = i-2$ since we know that we computed $F(i)$ using $F(i) = c_i + F(i-2)$
- if $F(i) = F(i-1)$, then we did not pick up coin c_i and we go back to $S(i) = i-1$ since we know that we computed $F(i)$ using $F(i) = F(i-1)$
- starting at coin 6:

  $F(6) > F(5)$, so we picked up coin 6 or $C_6 = 1$
  – we go to C_4 now
  $S(i) = 6-2 = 4$

  $F(4) > F(3)$, so we picked up coin 4 or $C_4 = 6$
  – we go to C_2 now
  $S(i) = 4-2 = 2$

  $F(2) > F(1)$, so we picked up coin 2 or $C_2 = 5$
  – we go to coin 0 or C_0
  $S(i) = 2-2 = 0$

- reached base-case. Stop here.

$$\boxed{\text{we pick up coins } C_6, C_4, \text{ and } C_2. \text{ This adds up to } 1+6+5 = 12}$$

4. Given the following items and a knapsack with weight capacity 5, find the most valuable subset of the items that fits in the knapsack. Fill out the table for F(i,j) and state the items in the knapsack. (10 points)

i	1	2	3	4
w_i	2	1	3	2
v_i	15	11	21	13

i\j	0	1	2	3	4	5
0						
1						
2						
3						
4						

i	1	2	3	4
W_i	2	1	3	2
V_i	15	11	21	13

i \ j	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	15	15	15	15
2	0	11	15	26	26	26
3	0	11	15	26	32	36
4	0	11	15	26	32	39

Recurrence Relation:

$$F(i,j) = \begin{cases} \max(v_i + F(i-1, j-w_i), F(i-1,j)), & w_i \le j \le W \\ F(i-1,j), & 0 \le j < w_i \end{cases}$$

Base case

$F(0,j) = 0$ with $0 \le j \le W$

$F(i,0) = 0$

F(1,1)

$0 \le j = 1 < w_1 = 2$

therefore $F(1,1) = F(0,1) = 0$

F(1,2)

$w_1 = 2 \le j = 2 \le W = 5$

and $V_1 = 15$, therefore

$F(1,2) = \max(15 + F(0,0), F(0,2))$

$= \max(15, 0) = 15$

F(1,3)

$w_1 = 2 \le j = 3 \le W = 5$ and $v_1 = 15$

therefore

$F(1,3) = \max(15 + F(1-1, 3-2), F(1-1, 3))$

$= \max(15 + 0, 0) = 15$

15 will also be the max for F(1,4) and F(1,5) since only 1 item in knapsack already included.

F(2,1)

$w_2 = 1 \le j = 1 \le W = 5$ and $V_2 = 11$

therefore

$F(2,1) = \max(11 + F(2-1, 1-1), F(2-1, 1))$

$= \max(11 + 0, 0) = 11$

F(2,2)

$W_2 = 1 \le j = 2 \le W = 5$ and $V_2 = 11$

therefore

$F(2,2) = \max(11 + F(2-1, 2-1), F(2-1, 2))$

$= \max(11 + 0, 15) = 15$

Here both items can be stored in the knapsack, but not at the same time, so we pick item with higher value ⇒ 15

F(2,3)

$W_2 = 1 \le j = 3 \le W = 5$ and $V_2 = 11$

therefore

$F(2,3) = \max(11 + F(2-1, 3-1), F(2-1, 3))$

$= \max(11 + 15, 15) = 26$

26 will also be the max for F(2,4) and F(2,5). Since items 1 and 2 already in knapsack

F(3,1)

We have $0 \leq j = 1 \leq w_3 = 3$ therefore

$F(3,1) = F(2,1) = 11$

F(3,2)

We have $0 \leq j = 2 < w_3 = 3$ therefore

$F(3,2) = F(2,2) = 15$

F(3,3)

$W_3 = 3 \leq j = 3 \leq W = 5$ and $V_3 = 21$

$F(3,3) = \max(21 + F(3-1, 3-3), F(3-1,3))$

$= \max(21 + 0, 26) = 26$

F(3,4)

$W_3 = 3 \leq j = 4 \leq W = 5$ and $V_3 = 21$

$F(3,4) = \max(21 + F(3-1, 4-3), F(3-1, 4))$

$= \max(21 + 11, 26) = 32$

F(3,5)

$W_3 = 3 \leq j = 5 \leq W = 5$ and $V_3 = 21$

$F(3,5) = \max(21 + F(3-1, 5-3), F(3-1, 5))$

$= \max((21 + 15), 26) = 36$

F(4,1)

We have $0 \leq j = 1 < W_4 = 2$

therefore $F(4,1) = F(3,1) = 11$

F(4,2)

$W_4 = 2 \leq j = 2 \leq W = 5$ and $V_4 = 13$

$F(4,2) = \max(13 + F(4-1, 2-2), F(4-1, 2))$

$= \max(13 + 0, 15) = 15$

F(4,3)

$W_4 = 2 \leq j = 3 \leq W = 5$ and $V_4 = 13$

$F(4,3) = \max(13 + F(4-1, 3-2), F(4-1, 3))$

$= \max(13 + 11, 26) = 26$

F(4,4)

$W_4 = 2 \leq j = 4 \leq W = 5$ and $V_4 = 13$

$F(4,4) = \max(13 + F(4-1, 4-2), F(4-1, 4))$

$= \max(13 + 15, 32) = 32$

F(4,5)

$W_4 = 2 \leq j = 5 \leq W = 5$ and $V_4 = 13$

$F(4,5) = \max(13 + F(4-1, 5-2), F(4-1, 5))$

$= \max(13 + 26, 36) = 39$

- Starting at $F(4,5)$

  Since $F(4,5) = 39 > F(3,5) = 36$ we picked up item 4 and we go back to $F(4-1, 5-2) = F(3,3)$

  Since $F(3,3) = F(2,3) = 26$ we did not pick up item 3 and go back to $F(2,3)$

  Since $F(2,3) = 26 > F(1,3) = 15$ we picked up item 2 and go back to $F(2-1, 3-1) = F(1,2)$

  Since $F(1,2) = 15 > F(0,2) = 0$ we picked up item 1 and we go back to $F(1-1, 2-2) = F(0,0)$

  ↳ Base case Stop here

$\boxed{\text{we pick up items } 4, 2, \text{ and } 1 \text{ for a value of } 39}$